

VILA REAL FLORES, Portugal

WMO: 085660

Lat: 41.317N

Lon: 7.733W

Elev: 1578

StdP: 13.88

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.8	30.9	22.6	18.0	34.3	25.3	20.4	34.8	18.0	49.9	15.8	46.9	1.3	100	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	92.1	71.2	88.6	69.1	85.5	67.6	72.4	90.1	70.3	86.6	68.3	83.5	4.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	100.2	83.7	63.9	94.4	79.7	62.5	89.5	76.9	36.9	90.8	35.1	86.9	33.4	83.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.5	11.9	10.1	DB	25.9	98.4	2.6	2.6	24.0	100.3	22.4	101.8	20.9	103.3	19.0	105.2
			WB	24.4	74.6	2.7	2.0	22.4	76.1	20.8	77.3	19.3	78.4	17.3	79.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	41.6	45.8	49.7	52.0	57.7	65.3	70.8	70.1	67.9	57.5	49.5	44.9	(6)
	DBStd	11.50	5.47	5.04	5.07	6.16	6.59	6.25	6.39	5.41	6.08	5.56	5.86	5.24	(7)
	HDD50	781	262	132	68	49	11	1	0	0	0	5	80	173	(8)
	HDD65	3839	725	539	473	391	241	72	18	14	37	242	465	622	(9)
	CDD50	3012	2	13	60	108	249	459	646	622	536	236	65	16	(10)
	CDD65	595	0	0	0	0	15	80	198	171	123	8	0	0	(11)
	CDH74	5979	0	0	1	4	169	802	2085	1732	1115	71	0	0	(12)
	CDH80	2174	0	0	0	0	23	231	878	653	384	5	0	0	(13)
(14) Wind	WSAvg	3.5	2.8	3.7	4.2	4.3	3.9	3.8	3.5	3.9	3.2	2.8	2.9	3.0	(14)
(15) Precipitation	PrecAvg	43.8	6.3	6.7	3.8	3.5	2.8	2.1	0.6	0.6	1.9	4.3	4.9	6.3	(15)
	PrecMax	71.3	17.5	17.7	14.1	10.4	7.5	9.4	2.9	2.7	6.4	10.8	18.1	24.4	(16)
	PrecMin	24.2	0.0	0.9	0.0	0.2	0.4	0.1	0.0	0.0	0.1	0.0	0.0	0.4	(17)
	PrecStd	11.6	4.4	5.0	3.5	2.6	1.8	2.1	0.8	0.7	1.6	3.1	3.9	5.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	61.9	70.7	74.3	83.3	89.6	96.9	97.0	92.9	79.8	64.9	60.9	(19)
		MCWB	50.9	53.7	57.7	58.7	66.9	69.8	74.5	73.7	70.8	65.2	58.3	56.2	(20)
	2%	DB	53.2	57.9	66.7	70.7	78.8	86.1	92.9	91.5	88.2	75.6	62.9	57.6	(21)
		MCWB	49.8	50.9	55.5	57.5	64.1	67.6	71.9	70.3	67.8	62.6	57.7	53.3	(22)
	5%	DB	51.5	55.2	62.9	67.5	74.6	82.4	89.1	87.3	84.5	71.6	60.2	54.9	(23)
		MCWB	48.5	50.3	53.0	55.8	61.8	65.5	69.9	68.2	66.2	60.6	55.7	51.4	(24)
	10%	DB	49.9	53.4	59.4	63.4	70.3	78.8	85.6	83.7	80.5	67.3	57.9	52.7	(25)
		MCWB	46.9	49.1	50.9	53.9	59.5	64.0	68.4	66.8	64.9	58.3	54.6	49.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	55.2	58.5	60.4	67.6	71.7	75.4	74.5	72.0	66.1	60.6	57.0	(27)
		MCDB	54.1	58.0	69.4	70.0	80.7	87.5	96.2	95.4	90.9	77.0	63.2	59.2	(28)
	2%	WB	50.8	53.5	56.1	58.6	64.5	68.5	72.9	71.6	68.7	63.2	58.2	53.8	(29)
		MCDB	52.4	56.0	65.5	69.0	77.6	83.9	90.8	89.4	86.0	73.6	61.4	56.6	(30)
	5%	WB	49.3	51.4	53.9	56.7	62.4	66.4	70.9	69.3	67.0	61.1	56.6	52.0	(31)
		MCDB	51.0	54.3	61.7	65.1	73.6	80.5	87.2	85.0	82.2	69.6	59.9	54.2	(32)
	10%	WB	47.3	49.7	51.9	54.7	60.1	64.6	68.8	67.3	65.3	59.3	54.9	50.1	(33)
		MCDB	49.3	52.7	57.7	62.0	69.1	77.4	84.0	81.6	78.6	66.2	57.5	52.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.9	16.0	15.3	18.4	20.3	23.2	24.3	21.8	15.9	10.6	9.8	(35)
		MCDBR	11.5	15.4	22.1	21.6	25.5	26.4	27.6	29.3	27.8	24.2	12.2	10.0	(36)
	5% WB	MCWBR	8.6	10.0	12.8	11.5	13.8	12.6	13.0	13.2	12.9	12.7	7.7	7.2	(37)
		MCWBR	9.1	12.2	20.2	19.9	23.9	25.2	26.4	27.8	26.1	21.3	10.1	8.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.303	0.315	0.347	0.351	0.360	0.362	0.349	0.362	0.350	0.340	0.312	0.300	(40)
		taud	2.538	2.493	2.416	2.409	2.408	2.414	2.432	2.391	2.449	2.498	2.537	2.542	(41)
	Ebn at Noon		273	285	286	292	290	289	291	285	281	272	266	264	(42)
		Edh at Noon	23	27	33	36	37	36	36	36	32	27	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		567	900	1243	1601	1987	2258	2354	2099	1606	1005	644	508	(44)
	RadStd		91	135	196	200	205	143	101	109	116	132	122	76	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page